

COS3302

Applied Data Science

Week 1

Introduction to Data Science

Agenda

- Overview of COS3302
- Introduction to Data Science
 - The difference between AI, ML, DL, and Data Science
 - Tools
- Assignment
- Question and Answer

Overview of COS3302

Monday: 1.30 P.M. to 5.20 P.M.

Class attendance:

Onsite: SCL211/2

Additional sources: Google classroom

Instructor

Dr.Chadaporn Keatmanee

- Ph.D. (Engineering and Technology), Sirindhorn International Institute of Technology (SIIT), Thammasat University, Thailand
- Ph.D. (Information Science), Japan Advanced Institute of Science and Technology (JAIST), Kanasawa, Japan
- M.Eng. (Computer Science), Asian Institute of Technology (AIT), Pathumthani, Thailand
- B.Id.Ed. (Telecommunication Engineering), King Mongkut's Institute of Technology Ladkrabang (KMITL), Thailand
- Email: chadaporn@rumail.ru.ac.th, chadaporn@ru.ac.th
- Room: Faculty rooms (opposite to SCL201)

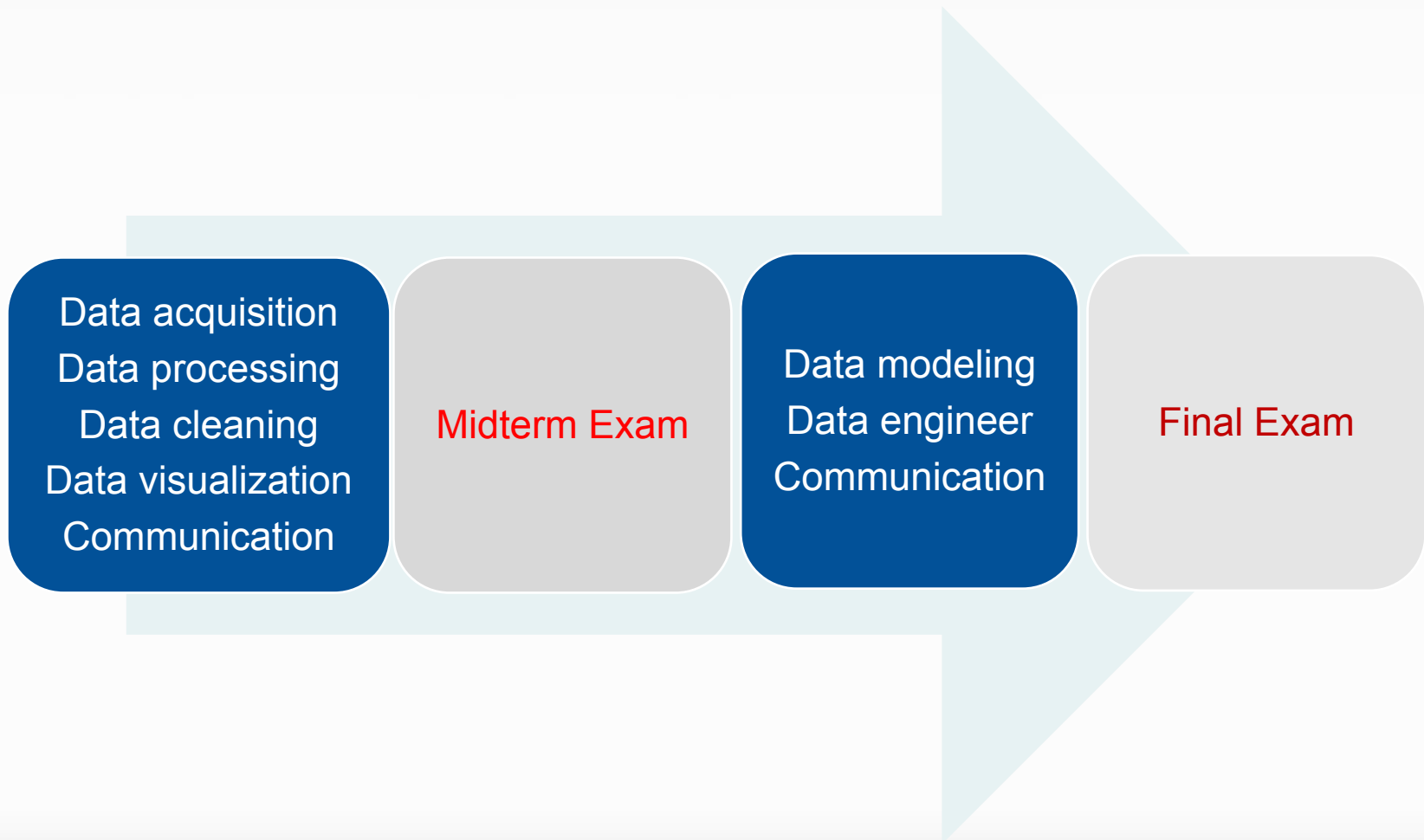
Teaching method

Lecture : 2 + 2 Hours

Theory

Workshop

Topic



Class Attitude

10 SIMPLE WAYS TO ENGAGE IN LIFELONG LEARNING

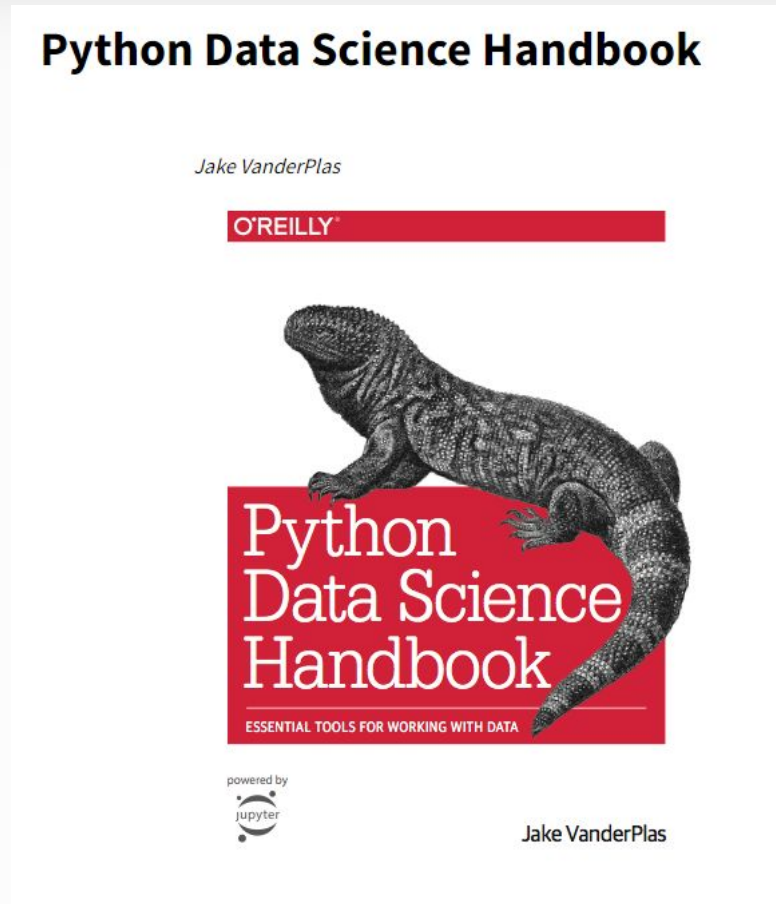
1. Read widely and often
2. Keep smart company
3. Teach others
4. Keep a list of things you want to explore
5. Start your own project
6. Use a personal learning environment
7. Experiment with new ways to learn
8. Join a study group
9. Find a job that encourages learning and collaboration
10. Make it a priority!

Evaluation

Score – 100%

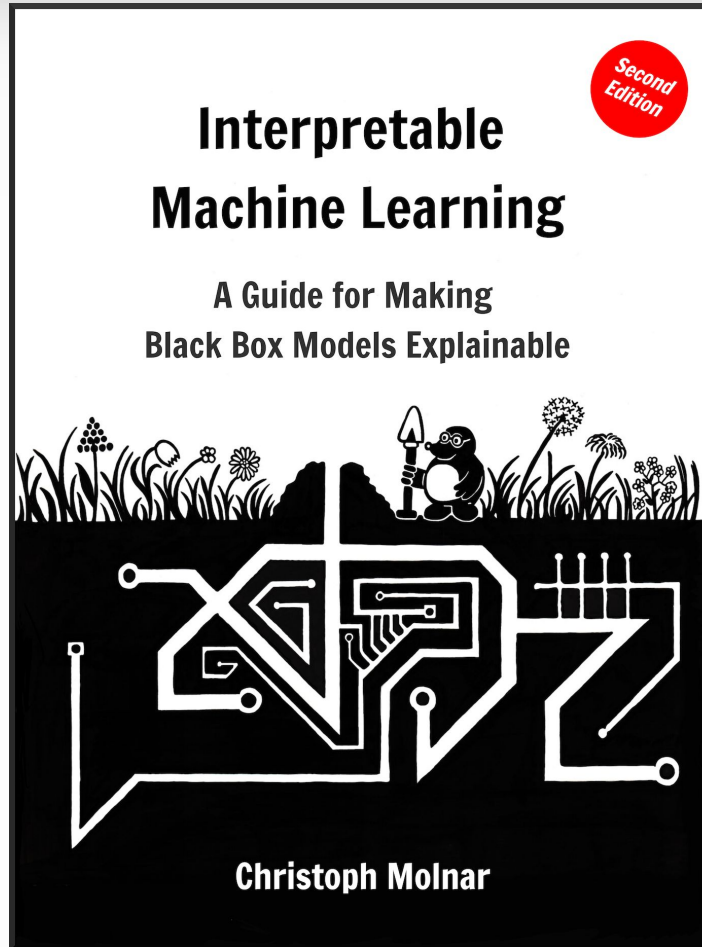
- Assignment 20%
- Midterm 40%
- Final 40%

Textbook



<https://jakevdp.github.io/PythonDataScienceHandbook/>

Textbook



<https://christophm.github.io/interpretable-ml-book/>

Resource

CamDavidsonPilon / Probabilistic-Programming-and-Bayesian-Methods-for-Hackers Public Sponsor

[Code](#) [Issues 147](#) [Pull requests 45](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#)

master 3 branches 0 tags Go to file Code

CamDavidsonPilon Create FUNDING.yml efc9398 on Mar 15 1,094 commits

.github	Create FUNDING.yml	4 months ago
Chapter1_Introduction	normed changed to density	9 months ago
Chapter2_MorePyMC	normed changed to density	9 months ago
Chapter3_MCMC	normed changed to density	9 months ago
Chapter4_TheGreatestTheoremNever...	normed changed to density	9 months ago
Chapter5_LossFunctions	normed changed to density	9 months ago
Chapter6_Priorities	normed changed to density	9 months ago
Chapter7_BayesianMachineLearning	docs: Fix a few typos	12 months ago

<https://github.com/CamDavidsonPilon/Probabilistic-Programming-and-Bayesian-Methods-for-Hackers>

Introduction of Data Science

What is Data Science?

- Data analysis
 - **The core is the business**
- Associate the analysis logic of the business with the processing logic of data analysis, and data analysis tools
 - **To achieve results**
- Tools; vehicles are chosen differently according to different roads

Identify the needs

- Identify the needs of analysts (in the enterprise):

- **Business analysts:**

- Marketing, sales, healthcare, banking, finance, policy work, etc.

- **Technology analysts:**

- IT department, Data center

Data Science

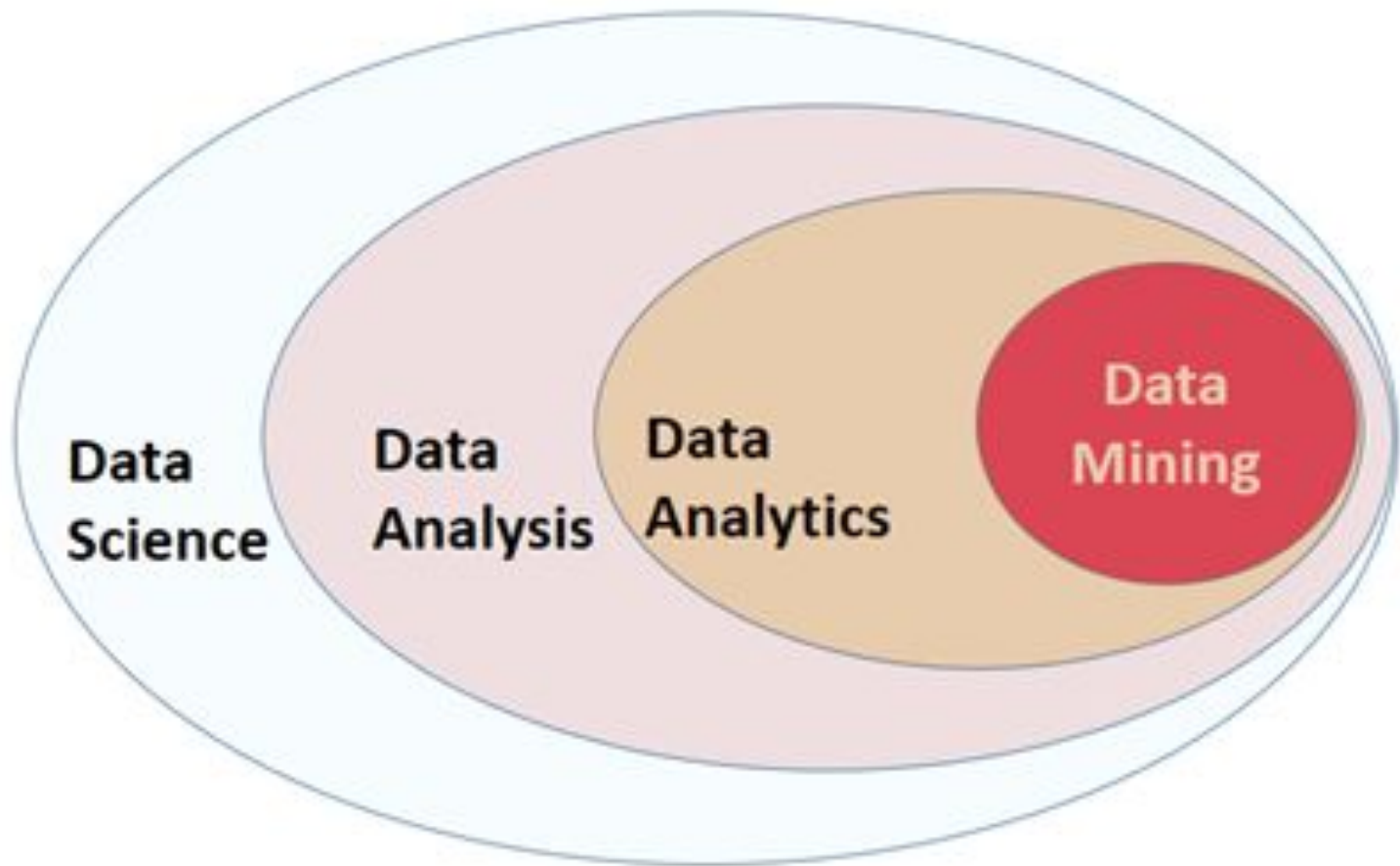
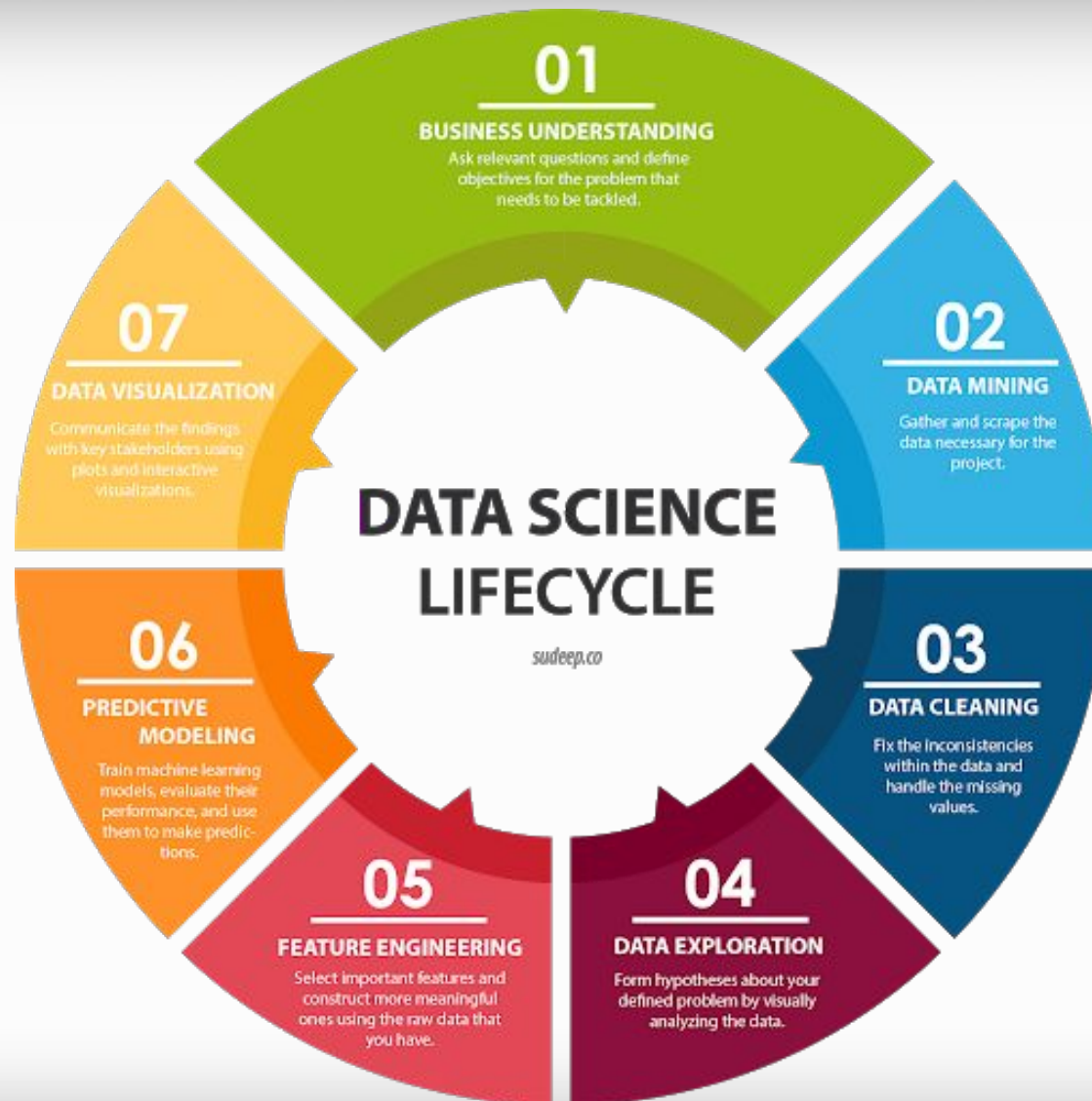
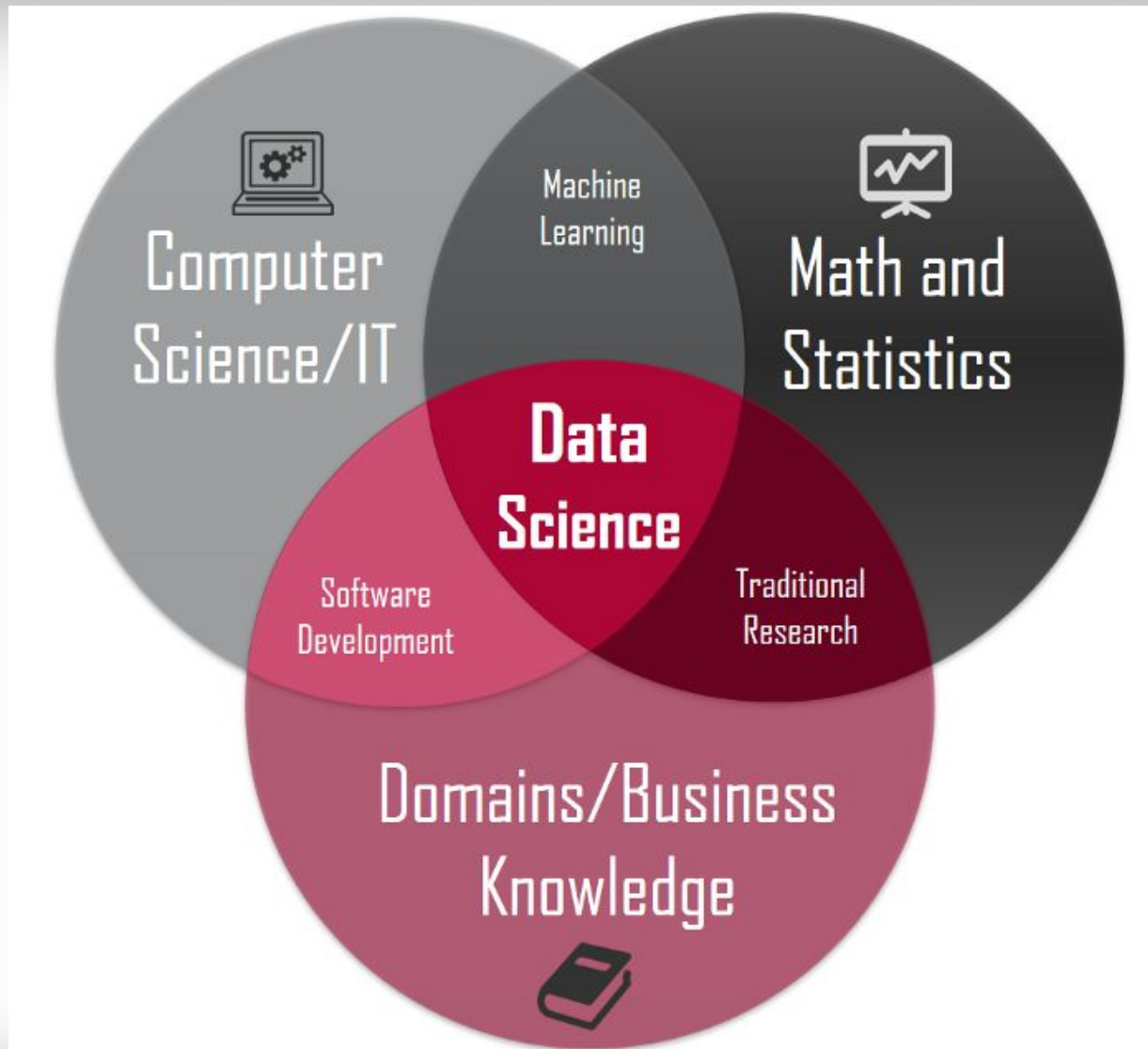


Image from Coursack

Tasks in Data Science:



Skills in Data Science:



AI vs Machine Learning vs Deep Learning

ARTIFICIAL INTELLIGENCE

Early artificial intelligence stirs excitement.



MACHINE LEARNING

Machine learning begins to flourish.



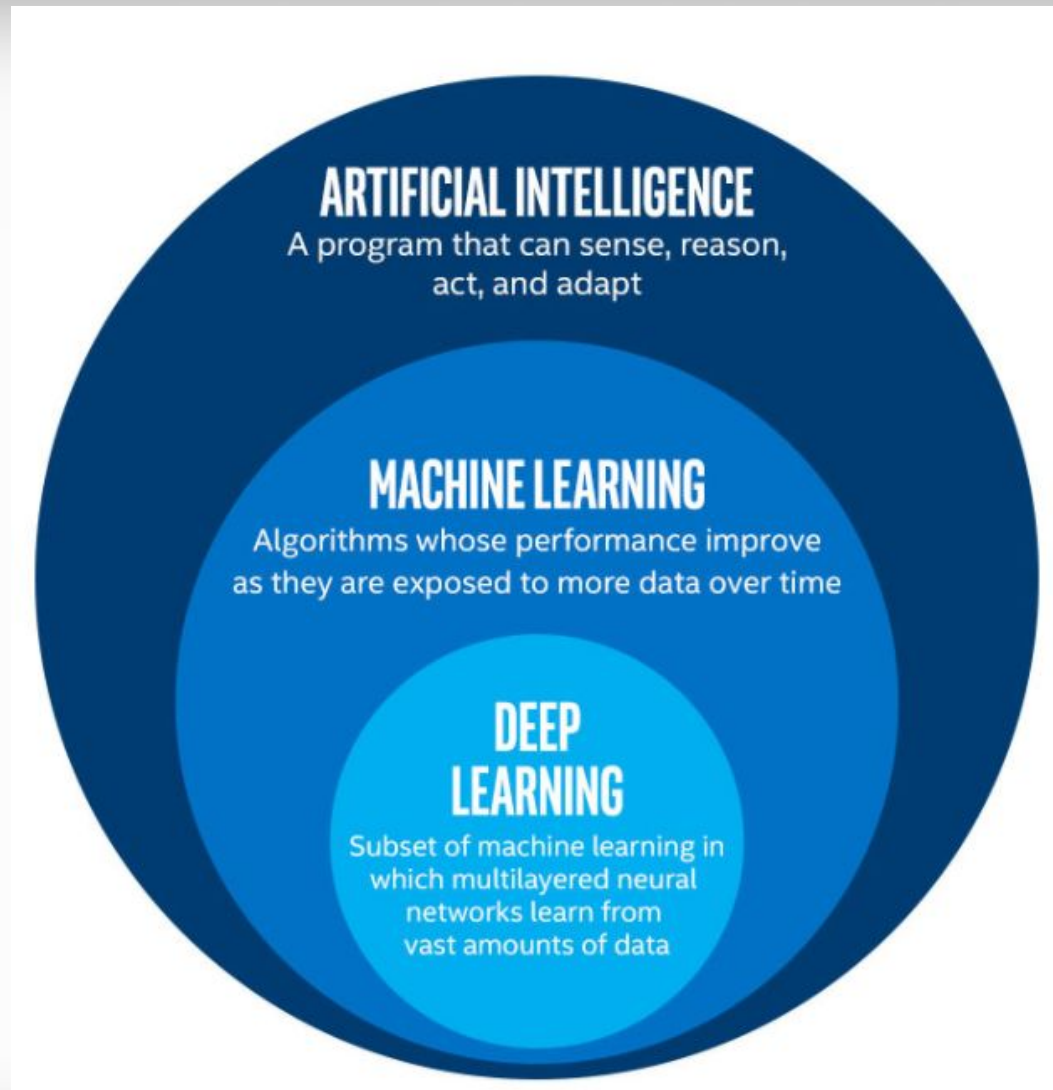
DEEP LEARNING

Deep learning breakthroughs drive AI boom.



Since an early flush of optimism in the 1950's, smaller subsets of artificial intelligence - first machine learning, then deep learning, a subset of machine learning - have created ever larger disruptions.

AI **vs** Machine Learning **vs** Deep Learning



Machine Learning **vs** Data Science

Machine learning vs. data science

Home
prices

size (square feet)	# of bedrooms	# of bathrooms	newly renovated	price (1000\$)
523	1	2	N	100
645	1	3	N	150
708	2	1	N	200
1034	3	3	Y	300
2290	4	4	N	350
2545	4	5	Y	440

Homes with 3 bedrooms are more expensive than homes with 2 bedrooms of a similar size.

Newly renovated homes have a 15% premium.

Running AI system
(e.g., websites / mobile app)

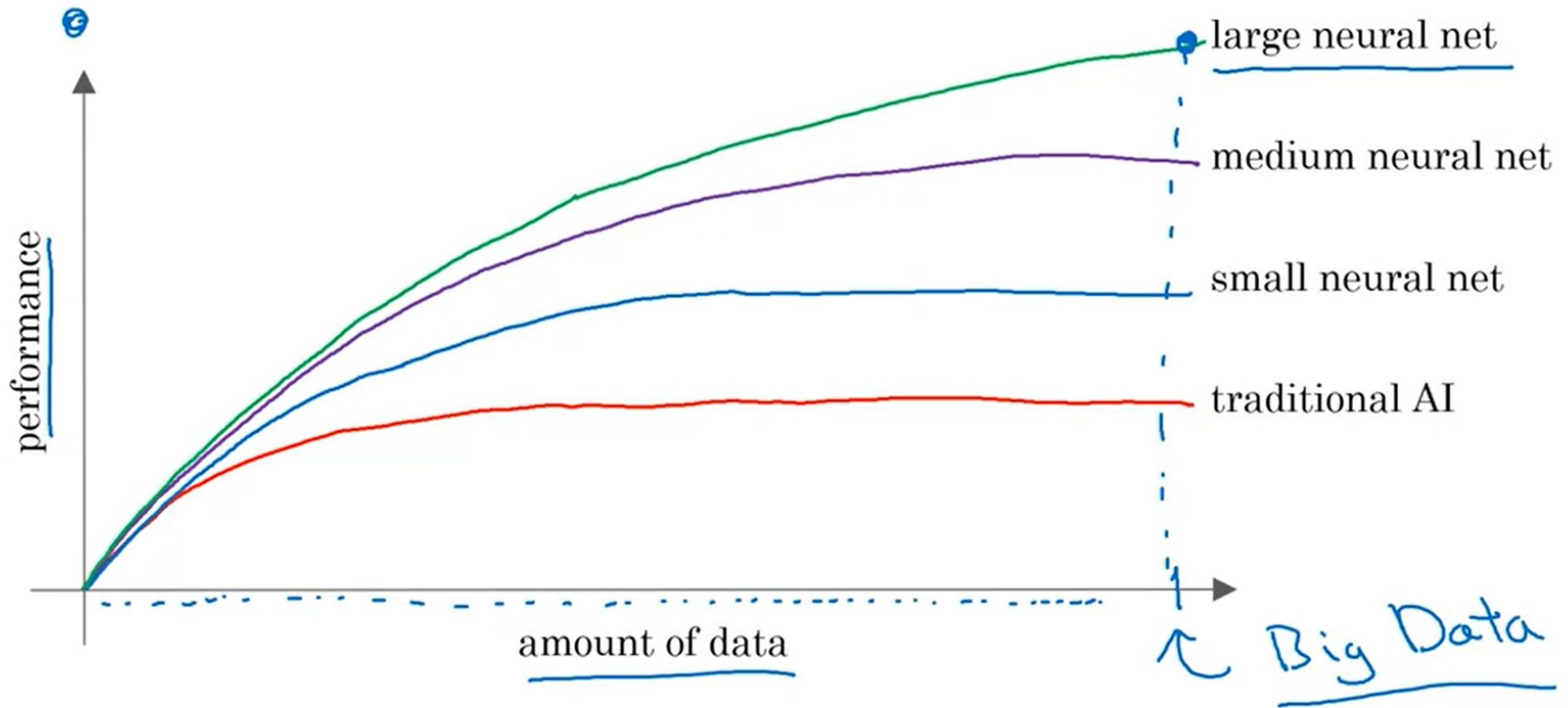
Supported algorithms for Data Science

AI has many tools

- Machine learning and data science
- Deep learning / neural network
- Other buzzwords: Unsupervised learning, graphical models, planning, knowledge graph, ...

Supported algorithms for Data Science

Why Now?



Identify Attributes of Tools:

- Analysis tools or code tools.



Identify Attributes of Tools:

Dimension 1:

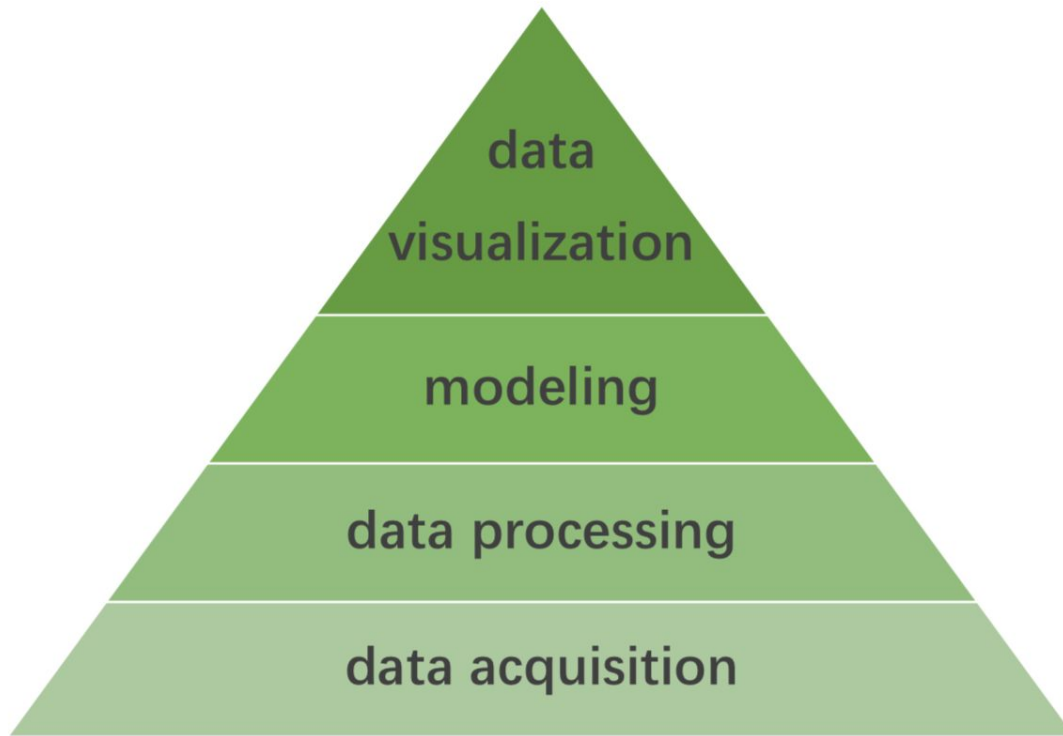
- Excel (Visual Basic Application: VBA)
- SPSS
- SAS
- **R (Data mining engineers)**
- Matlab
- **Python (Data mining engineers)**

Identify Attributes of Tools:

Code tools:

- SQL
- Hadoop
- Hive
- Shell
- Java

Identify Attributes of Tools:



Offline interface: Excel, Powerpoint, Tableau
Offline code: R, SAS, Python, D3.js, Process
Online: Echarts, Tagxedo

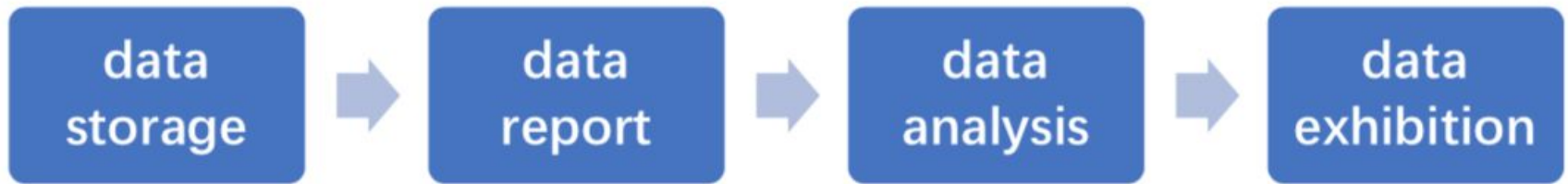
Dedicated interface: SPSS
Dedicated code: R, SAS
Generic code: Python

Database: SQL, Hive
Interface: Excel, SPSS
Code: VBA, Python, R, SAS

Database: SQL, Hadoop, Hive
Crawler: Python, Java

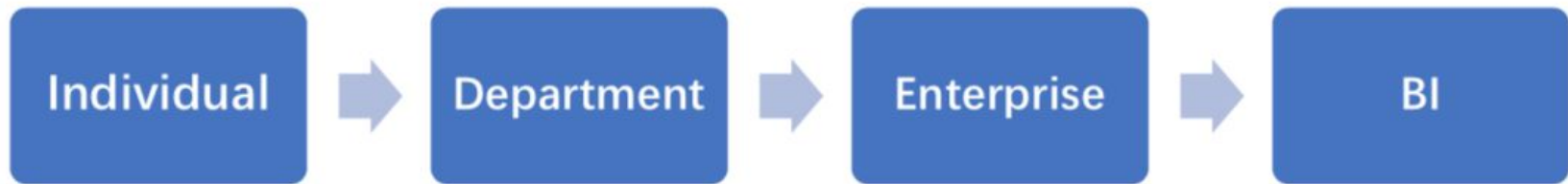
Identify the data architecture of the enterprise:

Dimension 1:



Identify the data architecture of the enterprise:

Dimension 2:



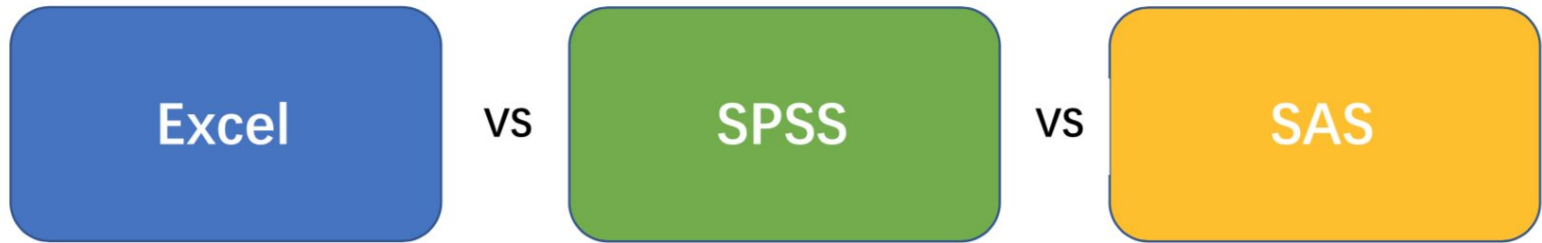
Data Storage

- **MySQL**
- **SQL Server**
- **DB2 (Enterprise)**
- **Oracle (Enterprise)**

Data Report

- **Tableau**
- **PowerBI**
- **FineReport**
- **Qlikview**

Data Analysis



<https://towardsdatascience.com/what-data-analysis-tools-should-i-learn-to-start-a-career-as-a-data-analyst-af650b54c9e9>

Data Visualization (Exhibition)

Daily States of SH Rail Transport

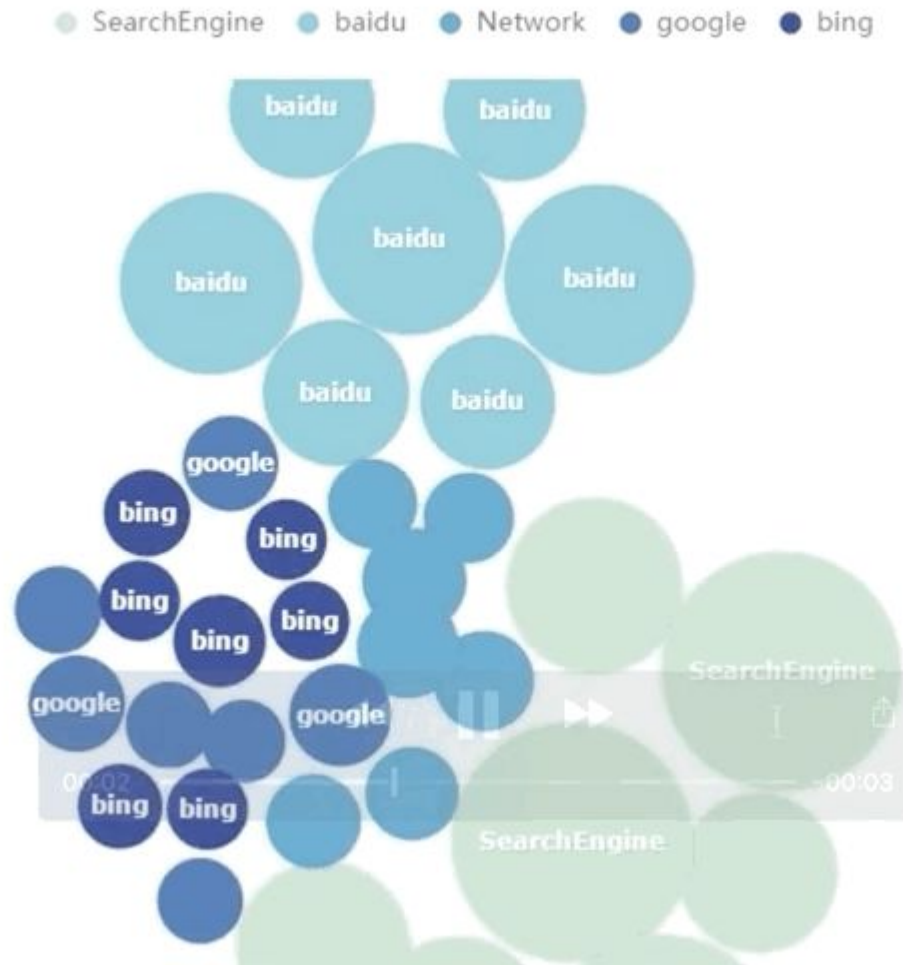
Transportation Revenue (Yuan)									
Today				Monthly				Yearly	
33974872.47				33974872.47				8914304128.66	
All lines		Train		Workload		Turnaround		Stopping	
Plan		10424		6949		1.5		11.5	
Actual		10180		7486		1.36		10.3	
Technical Station States									
South Station									
Serviceable Train			Loaded Train	Empty Train	Weight (tons)	Stopping		Mean-time	
Plan	SH	1050			451667	20	H	1.5	H
Actual	SH	796	317	479	480586		H	1.3	H
Full Site	935	450	485						
Disintegrated			Marshalling g	Handling Train					
	Train		Train	488	Train	32364		Vehicle	
North Station									
Serviceable Train			Loaded Train	Empty Train	Stopping		Mean-time		
Plan	900				2	H	1.4	H	
Actual	474		132	342	5	H	1.2	H	
Weight (tons)			Rejection Weight		Fill Empty		Handling Train		
Plan	406287		13	Train	16	Train	433	Train	
Actual	460888		137	Vehicle	127	Vehicle	28579	Vehicle	

Train Access to Each Station								
Entrance & Exit		number				number		
		details	details	details	details	details	details	details
daily	Entrance	14	64	4707	318485		78	4762
	Entrance		13	45864	45864		13	728
	Entrance		2	6552	6552		2	104
	Entrance		10	41580	41580		10	653
	Entrance		5	20526	20526		6	396
	Entrance		22	91528	91528		21	1383
	Total	14	116	210757	524535	0	130	8026
monthly	Entrance	14	64	4707	318485		78	4762
	Entrance		13	728	45864		13	728
	Entrance		2	104	6552		2	104
	Entrance		10	659	41580		10	653
	Entrance		5	330	20526		6	396
	Entrance		22	1448	91528			
	Total	14	116	7976	524535	0	109	6643
yearly	Entrance	2341	16773	1177943	78710243		19132	1176817
	Entrance		3022	169206	10659962		3016	168643
	Entrance		621	32292	2034390		621	32289
	Entrance		4696	306166	1784818		4693	305936
	Entrance		1012	66700	4152415		924	60918
	Entrance		4805	316564	20072754			
	Total	2341	30929	2068871	117414582	0	28386	1744603

Data Visualization (Exhibition)



Data Visualization (Exhibition)



Data Visualization (Exhibition)

Comparison of Data Analysis Tools:

- ☐ **Excel**
- ☐ **R**
- ☐ **Python**
- ☐ **BI Tools**

Excel

Usage scenarios

- Data processing work under general office requirements
- Data management and storage of small and medium-sized companies
- Simple statistical analysis for students or teachers (such as analysis of variance, regression analysis, etc.)
- Combine Word and PowerPoint to create data analysis reports

Excel

Usage scenarios

- Assistant tool of data analysts
- Production of charts for some business magazines and newspapers and newspaper (data visualization)

Excel

Usage scenarios

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- Production of charts for some business magazines and newspapers and newspaper (data visualization)

Excel

Advantages

- It's easy to get started with Excel
- The learning resources are very rich
- You can do a lot of things with Excel: modeling, visualization, reports, dynamic charts, etc.
- It can help you understand the meaning of many operations before further learning other tools (such as Python and R)

Excel

Disadvantages

- To fully master Excel, you need to learn VBA (visual basic application), so the difficulty is still very high
- When the amount of data is large, there will be a situation of stuttering
- The excel data file itself can hold only 1.08 million rows without the aid of other tools, and it's not suitable for processing large-scale data sets

Excel

Disadvantages

- The built-in statistical analysis is too simple and has little practical value
- Unlike Python, R, and other open source software, there is a charge for the genuine Excel

R

Usage scenarios

- Data cleaning and data reduction
- Web crawling
- Statistical hypothesis testing (t test, analysis of variance, chi-square test, etc.
- Statistical modeling (linear regression, logistic regression, tree model, neural network, etc.
- Data analysis report output (R markdown)

BI

- The text is not as good as the table, and the table is not as good as the graph
- Data visualization is one of the main directions of data analysis
- Business Intelligent (BI) is that it is better at interactions and reporting.
- It is good at interpreting both historical and real-time data

Python

Usage scenarios

- Data crawling
- Data cleaning
- Construct data analysis algorithms based on the business scenario and actual problems
- Data visualization
- Data modeling
- Advanced fields of data mining and analysis, such as machine learning and text mining

Workshop

❖ Study applied data science 5 industries from 10 industries listed below;

1. Healthcare
2. Financial service
3. Gaming, sport, and entertainment
4. Energy, oil, and gas
5. Aerospace
6. Manufacturing, logistics, and supply chain
7. Insurance
8. Transportation and travel
9. Management consulting and customer service
10. Retail and e-commerce

Assignment

- ❖ Create a presentation between 10 to 15 slides
- ❖ Present next week (Monday November 24, 2025)
- ❖ Upload the following item in Google Classroom
 - Presentation file

References

- Coursera: AI for Everyone
- <https://csweb.rice.edu/academics/graduate-programs/online-mds/blog/data-science-industry-applications>

Question and Answer



Further information

& discussion: Email

Line group

Google classroom